Friction Before Failure: Making Review Burden Visible in Regulated Document Work

SORA KIM, Pharmaceutical Manufacturing Industry, South Korea

Deviation records and Corrective and Preventive Action (CAPA) documents in pharmaceutical manufacturing environments are reviewed by multiple actors across iterative revision cycles and formal approval chains — yet the coordination work that produces delay remains invisible to any single participant. This paper argues that review friction is not only a workflow inefficiency but an HCI problem about how interaction conditions shape what coordination work becomes visible and actionable before failure occurs. I introduce a judgment-preserving observability layer that surfaces hidden coordination work without automating quality decisions. I report a two-part prototype-based study with $N = 20$ QA/QC-experienced participants: a six-tab workflow dashboard that surfaces revision burden, comment distribution, and document journey patterns as interpretable signals, and a TF-IDF pattern detection layer that identifies recurring issue clusters across review cycles. Neither component modifies validated documentation systems. Participants consistently rated both components highly and expressed adoption intent, while reporting that friction visibility enabled them to interpret delay as a structural condition rather than an individual failure. The study's central empirical finding is a dissociation between immediate legibility and final endorsement: the pre-review checklist was immediately accessible because it addresses a problem practitioners already recognize, while the pattern detection layer required initial conceptual orientation because it addresses a problem practitioners have not previously had the tools to name — structural friction accumulation. Both arrived at equivalent ratings. I introduce the judgment-preservation constraint as a novel design boundary for workflow support research in regulated environments, and argue that making hidden coordination work legible is a precondition for more equitable accountability in regulated sociotechnical organizations.

Additional Key Words and Phrases: review friction, workflow visibility, CAPA, pharmaceutical manufacturing, GMP, invisible work, judgment-preserving design, regulated environments

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1 Introduction

In pharmaceutical manufacturing environments governed by Good Manufacturing Practice (GMP) regulations, deviation and corrective action (CAPA) records are among the most consequential documents a quality organization produces. Their review involves multiple actors — quality assurance specialists, subject matter experts, department managers — cycling through iterative rounds of comment, revision, and approval. These review processes are routinely experienced as slow. But the standard diagnosis of that slowness — too many reviewers, unclear ownership, insufficient training — misidentifies the problem. The core issue is not delay itself. It is that the coordination work producing the delay is structurally invisible.

Author's Contact Information: Sora Kim, Pharmaceutical Manufacturing Industry, South Korea, sorakim.lab@gmail.com.

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Review friction accumulates gradually: a wording comment triggers a revision cycle; a reviewer reopens a document closed in a previous round; a pattern of minor but recurring feedback generates disproportionate turnaround time across a document class. These patterns are real, consequential, and structural — but they remain distributed across individual comment threads, document versions, and reviewer roles. No single actor has a complete view of them. As a result, friction becomes visible only after delays have already escalated, and accountability is assigned retroactively, without access to the coordination conditions that produced the outcome. This is not only a workflow problem. It is an interaction problem: the system does not make coordination work visible at the moment it could be interpreted and acted upon.

This paper introduces a design intervention that addresses this observability problem directly: an external workflow visibility layer that surfaces review friction as interpretable signal without modifying validated documentation systems or automating quality decisions. The distinction matters: in GMP environments, human interpretive authority over quality decisions is not merely a design preference but a regulatory requirement. The design challenge is therefore to make friction visible while leaving judgment intact — the judgment-preservation constraint that defines the core design boundary of this work.

I report a two-part prototype-based study. In the first part, I developed a six-tab workflow dashboard over a synthetic dataset of 100 deviation-CAPA review records, and evaluated it with $N = 20$ QA/QC-experienced participants. In the second part, I added a TF-IDF pattern detection layer to surface recurring issue patterns across review cycles. Neither component modifies validated documentation systems; both operate as external observability layers.

The contributions of this work are threefold. First, I introduce review friction visibility as a design problem specific to regulated document environments, identifying the judgment-preservation constraint as a novel design boundary for workflow support research. Second, I demonstrate that a lightweight observability layer can surface hidden coordination work without automating quality decisions. Third, I offer a theoretical framing that connects HCI concerns about workflow visibility to broader questions about how accountability is structured in regulated sociotechnical organizations.

2 Related Work

Awareness and Coordination in Collaborative Work Systems

A foundational concern in CSCW research is how participants in distributed collaborative work maintain awareness of each other's activities. Dourish and Bellotti [2] establish that awareness — understanding of the activities of others that provides context for one's own action — is critical to successful collaboration and is poorly served by active, explicit information-sharing mechanisms. Their study of shared editing systems demonstrates that passively available awareness information, embedded in the shared workspace itself, allows participants to coordinate dynamically without the overhead of explicit communication. Applied to regulated document review, this framework implies that the coordination work occurring across review cycles — who has commented, which rounds are open, where revision burden is accumulating — should ideally be available as ambient, interpretable signal rather than requiring active compilation by individual reviewers.

Invisible Work and the Limits of Formal Indicators

Coordination work in distributed systems often becomes invisible precisely because it is effective: when practitioners manage revision loops, absorb comment burden, and route documents through informal channels, the effort expended does not appear in any formal record. Star and Strauss [6] theorize this as invisible work — labor that is performed but

not captured by organizational indicators, and therefore not attributed to the structural conditions that produce it. In regulated review environments, this invisibility has specific implications: the coordination work that generates delay sits outside CAPA records, audit trails, and quality metrics. The observability layer developed in this study makes this invisible work legible — not by automating its interpretation, but by surfacing the patterns that indicate its presence.

Situated Action and the Conditions of Review Work

Suchman's (1987) account of situated action establishes that action is not plan-driven but shaped by the immediate conditions of its execution — what is observable, reachable, and interpretable at the moment a decision must be made. Applied to document review, this implies that the quality of review decisions is shaped not only by reviewer expertise but by the conditions of visibility under which those decisions are made. A reviewer who cannot see how a document has moved through prior rounds, what patterns of comment have accumulated, or where the current cycle sits relative to historical revision burden is making decisions in conditions of reduced observability.

Human-AI Interaction and Compliance Constraints

The pattern detection layer in this study uses TF-IDF similarity clustering to surface recurring issue patterns across review cycles. The design of this layer is governed by a principle that Amershi et al. [1] formalize across guidelines for human-AI interaction: AI systems should make clear what they can and cannot do, display contextually relevant information, and support human control rather than supplant it. In regulated manufacturing environments, this principle is not merely a design preference but a compliance requirement: quality decisions must be made and documented by qualified human reviewers, and any system that automates or pre-empts those decisions introduces a regulatory risk.

Prototype-Based Evaluation

The methodological approach in this study builds on established HCI practice for evaluating prototype systems. Nielsen [5] demonstrates that think-aloud evaluation with small participant samples reliably surfaces usability problems and can be combined with post-task assessment to understand both navigational behavior and perceived utility. Hutchinson et al. [4] establish the logic of using a prototype simultaneously as a design intervention and as an analytic probe — a technology probe makes visible the conditions of use it is designed to address, rather than simply testing whether a given interface is usable.

AI-Assisted Decision Making and Judgment-Preserving Design

Within HCI, work on AI-assisted decision making has established that interface structure actively shapes which information becomes salient to decision makers. Wang and Yin [8] demonstrate that explanation interfaces alter how decision makers engage with outputs, with effects contingent on domain expertise — findings that establish observability tools as active shapers of interpretation rather than neutral displays. Green and Chen [3] show that algorithm-in-the-loop settings systematically constrain the conditions under which independent judgment is possible, establishing that the structural conditions of human-algorithm interaction determine whether human interpretive authority is genuinely preserved. In regulated environments where judgment preservation is a compliance requirement rather than a design preference, these findings ground the observability layer's design logic: surfacing coordination patterns without recommending actions preserves the interpretive authority that GMP regulation requires. Prior work has not examined review friction visibility as a design problem in regulated document environments, where the judgment-preservation constraint uniquely restricts what computational support is permissible. This study addresses that gap directly.

3 Method

Study Context

This study was conducted in the context of pharmaceutical manufacturing quality systems operating under GMP regulations. Deviation and CAPA review workflows in these environments involve multiple reviewers, iterative revision cycles, and formal approval chains — all of which generate coordination work that is rarely visible as a unified pattern to any individual participant.

Dataset Construction

The study used a synthetic dataset of 100 deviation-CAPA review records generated programmatically in Python. Because actual facility records are subject to non-disclosure obligations, the dataset was constructed to reflect the structural and terminological characteristics of real GMP review workflows — including comment categories, review round sequences, severity classifications, turnaround times, document reopening events, and reviewer role assignments — without reproducing proprietary content. The judgment-preservation constraint applies equally to synthetic and live records: observability tools must operate externally over data rather than within the document management system. This construction approach reflects a domain constraint rather than a methodological preference: GMP deviation and CAPA records are proprietary quality documents protected under confidentiality obligations that preclude their use in external research without facility authorization. Synthetic generation from structural parameters is the standard approach for research in regulated manufacturing contexts where live data access is unavailable.

Phase 1: Workflow Dashboard Prototype

Based on the synthetic dataset, I developed a six-tab workflow dashboard as a Streamlit web application (Figures 1–6). The dashboard was designed as an external observability layer — it reads from the dataset but does not modify, version, or interact with any validated documentation system. The six tabs surface distinct dimensions of review burden:

- **Overview** (Figure 1): comment category distribution, severity breakdown, average turnaround by reviewer role, and review round distribution across the synthetic dataset.
- **Heatmap** (Figure 2): revision bottleneck visualization across reviewer–document type combinations, with revision loop frequency indicators.
- **Checklist** (Figure 3): a pre-review scaffold with a rule-based comment classifier to support consistent review preparation, and a comment category × average turnaround chart.
- **Document Journey** (Figure 4): full review lifecycle trace with round-by-round summary for individual records.
- **Before/After** (Figure 5): comparative workflow support visualization showing anticipated changes across key dimensions.
- **Reviewer Simulation** (Figure 6): comment framing tool with friction estimation to support reflection on review patterns.

The design of each tab was governed by the judgment-preservation principle: the dashboard surfaces patterns and distributions as interpretable signals, but does not recommend actions, assign responsibility, or produce outputs that function as quality decisions.

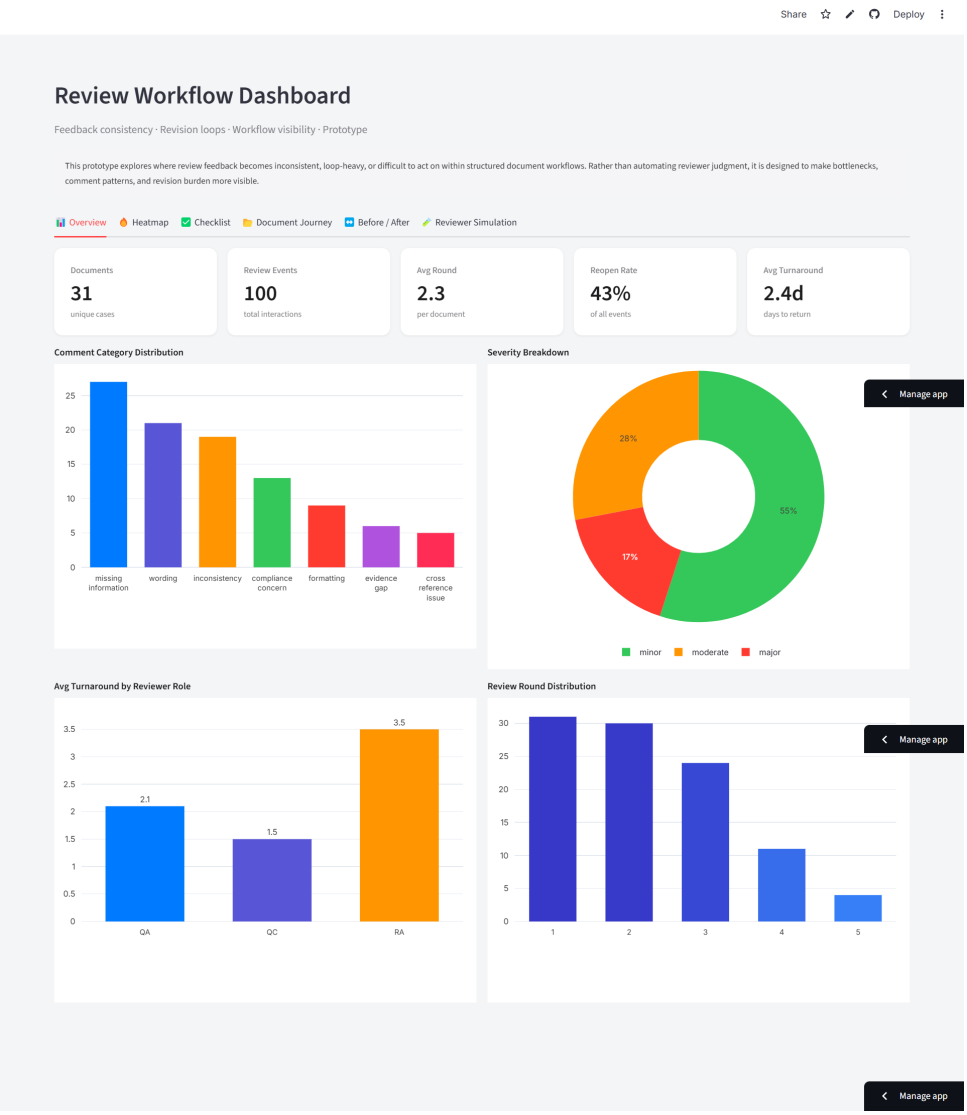


Fig. 1. Overview tab of the workflow dashboard, showing comment category distribution across the 100-record synthetic dataset, severity breakdown (55% minor, 28% moderate, 17% major), average turnaround by reviewer role (QA: 2.1d, QC: 1.5d, RA: 3.5d), and review round distribution. The RA role's substantially higher turnaround is visible as a pattern

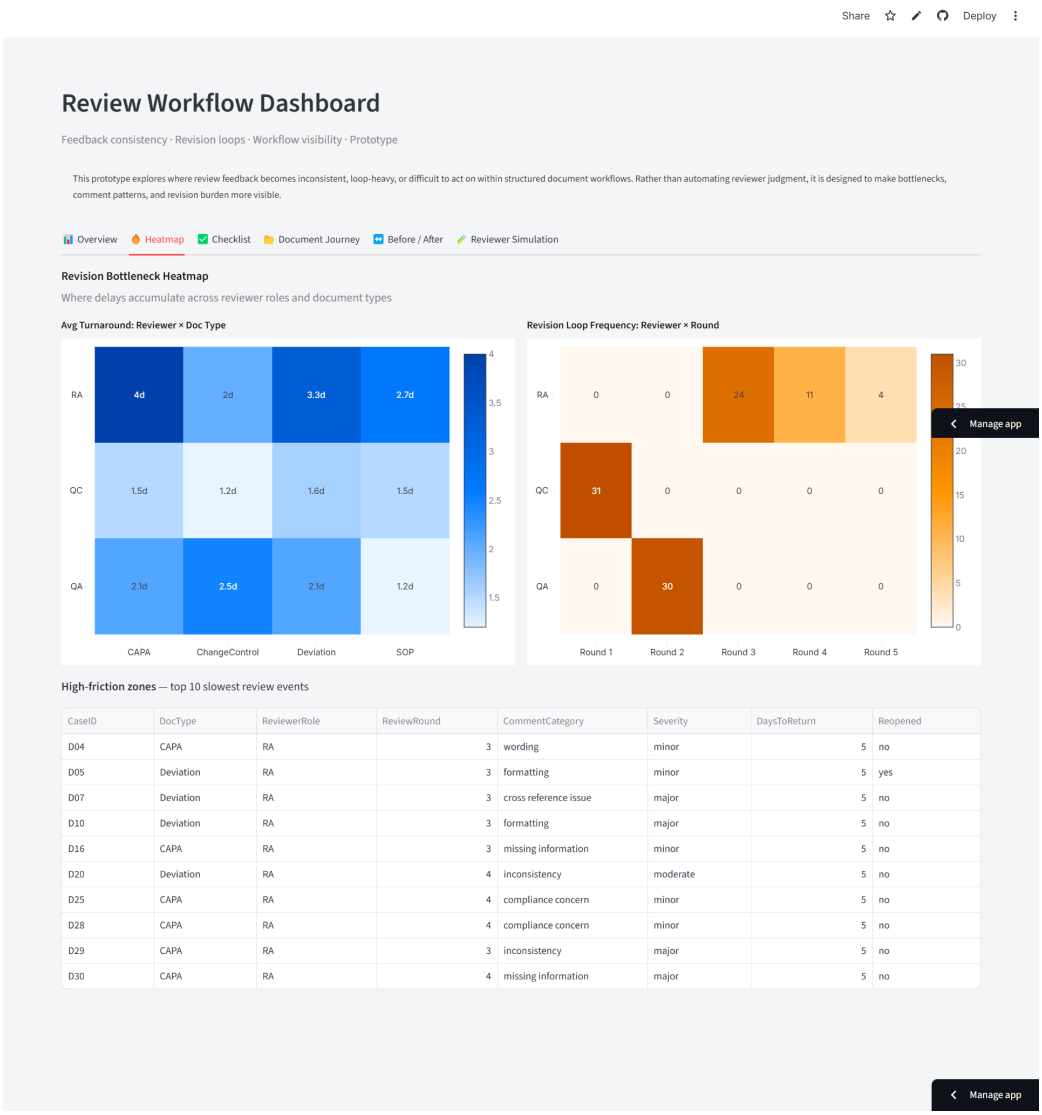


Fig. 2. Heatmap tab, showing average turnaround by reviewer role and document type (left) and revision loop frequency by reviewer and round (right). QC loops concentrate in Round 1 (31 events), QA loops in Round 2 (30 events), and RA loops spread across Rounds 3–5 — a structural pattern indicating that different reviewer roles generate friction at systematically different stages of the review lifecycle.

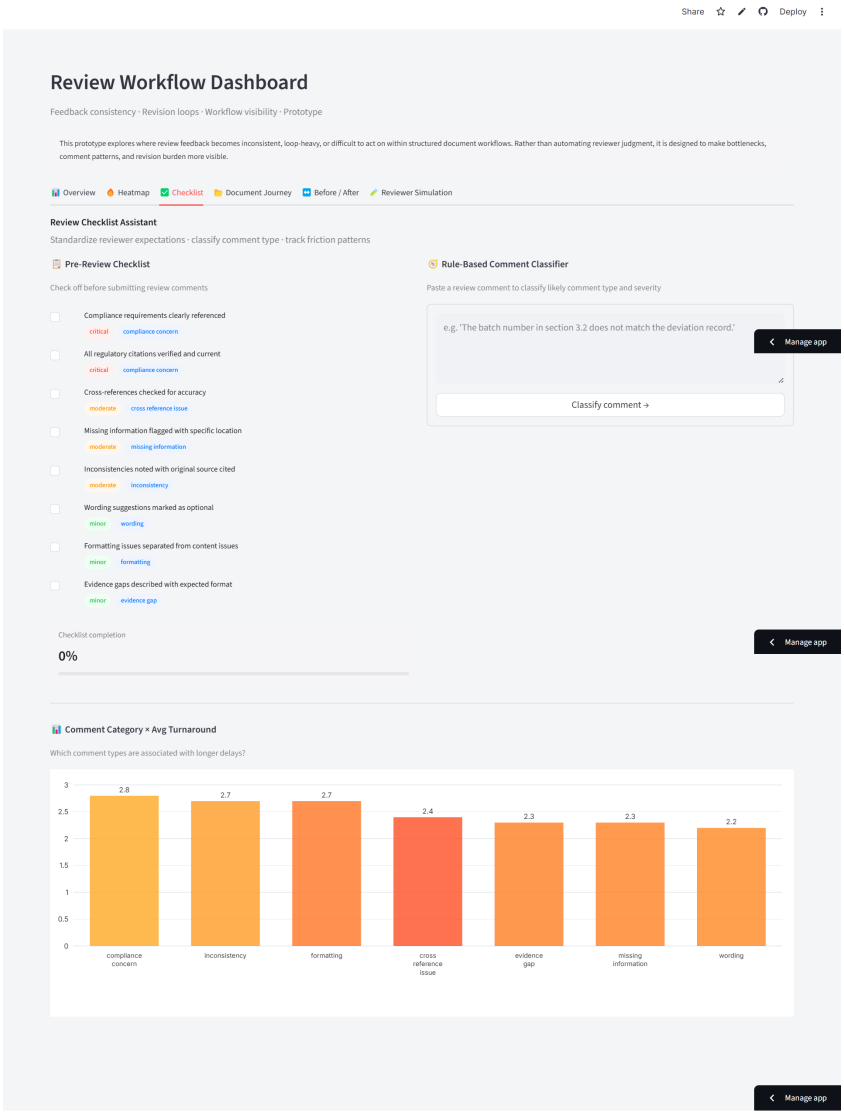


Fig. 3. Checklist tab, showing the pre-review checklist scaffold (left) and rule-based comment classifier (right). The checklist was the most immediately legible component in the pilot evaluation — rated 5.0/5 without requiring conceptual reorientation.

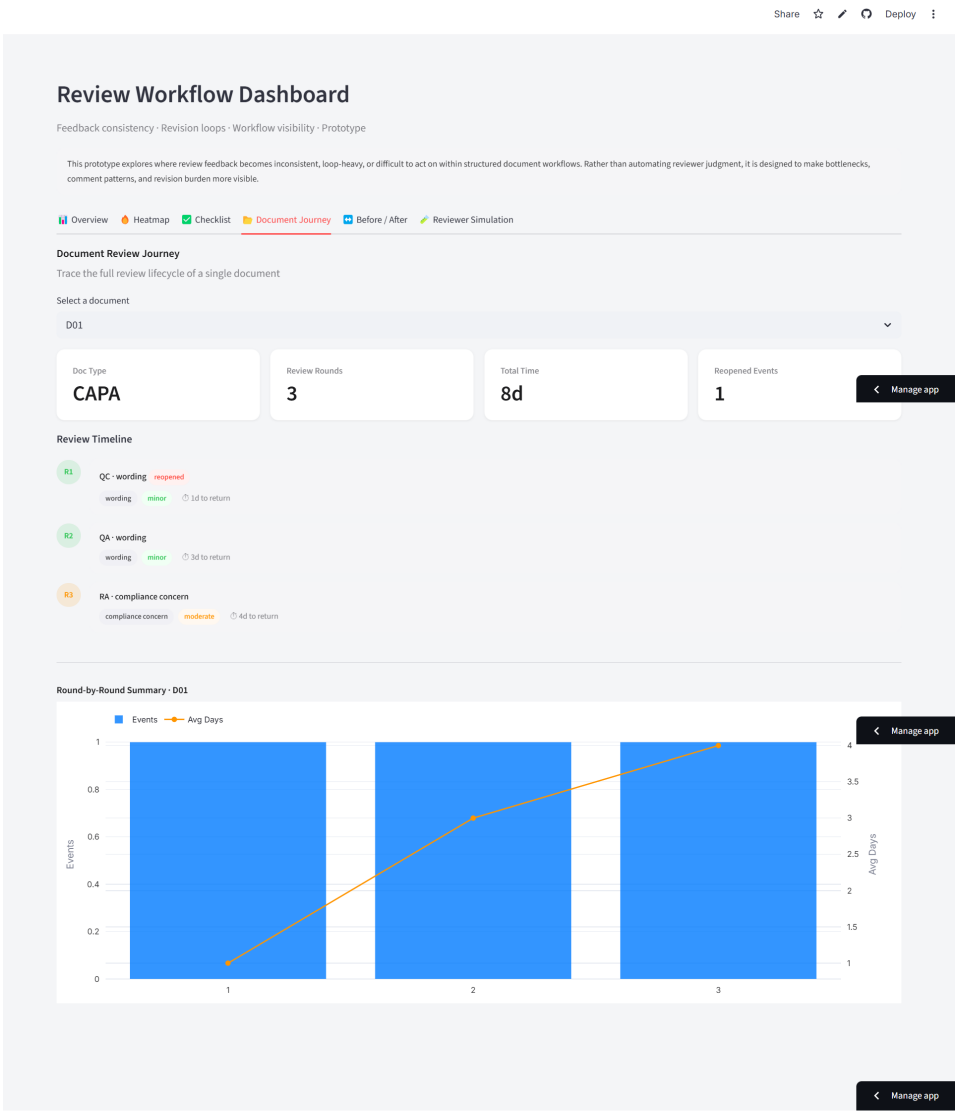


Fig. 4. Document Journey tab, showing the full review lifecycle of a single document (D01: CAPA, 3 review rounds, 8 days total, 1 reopened event). The review timeline shows round-by-round activity with comment category, severity, and return time for each event. The round-by-round summary chart shows escalating turnaround across rounds

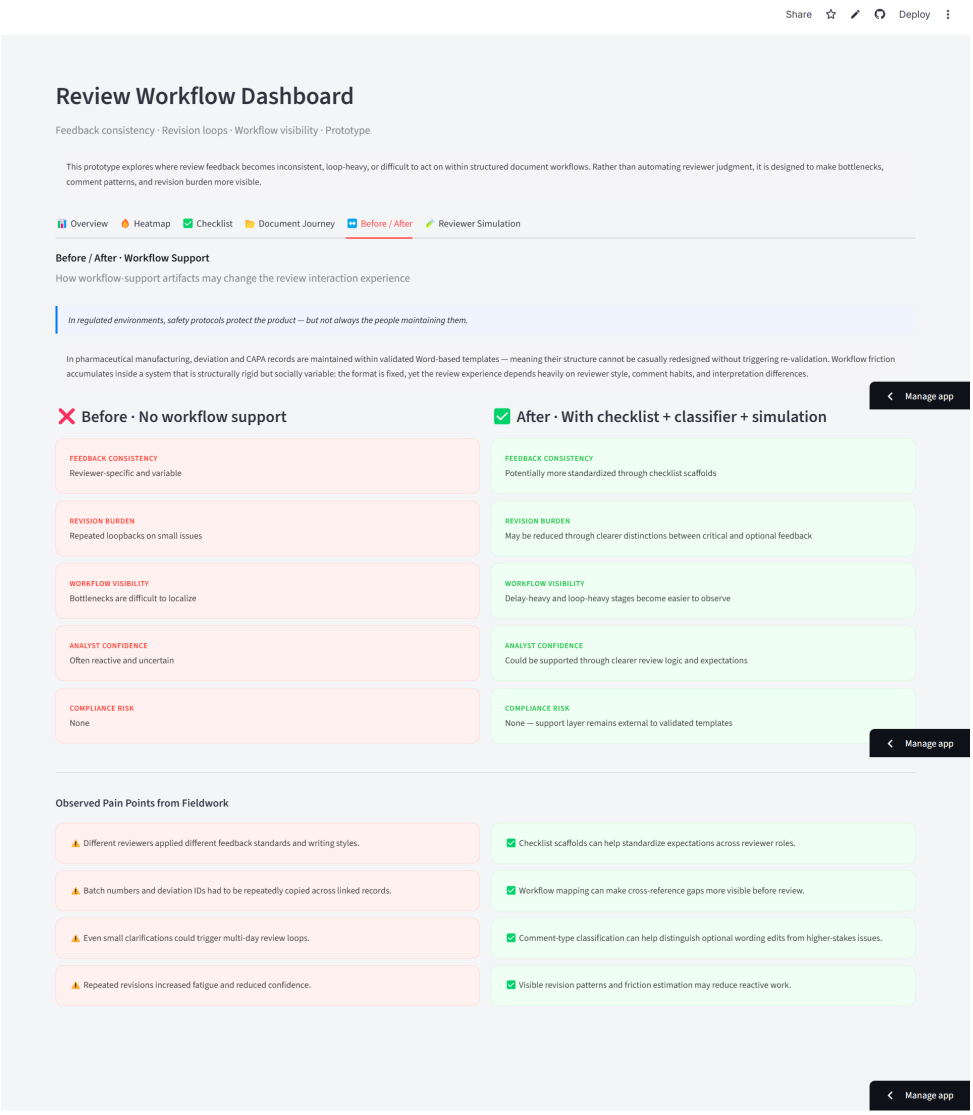


Fig. 5. Before/After tab, showing anticipated changes across five dimensions (feedback consistency, revision burden, workflow visibility, analyst confidence, compliance risk) when the observability layer is applied. Observed pain points from fieldwork are listed below, with corresponding design responses.

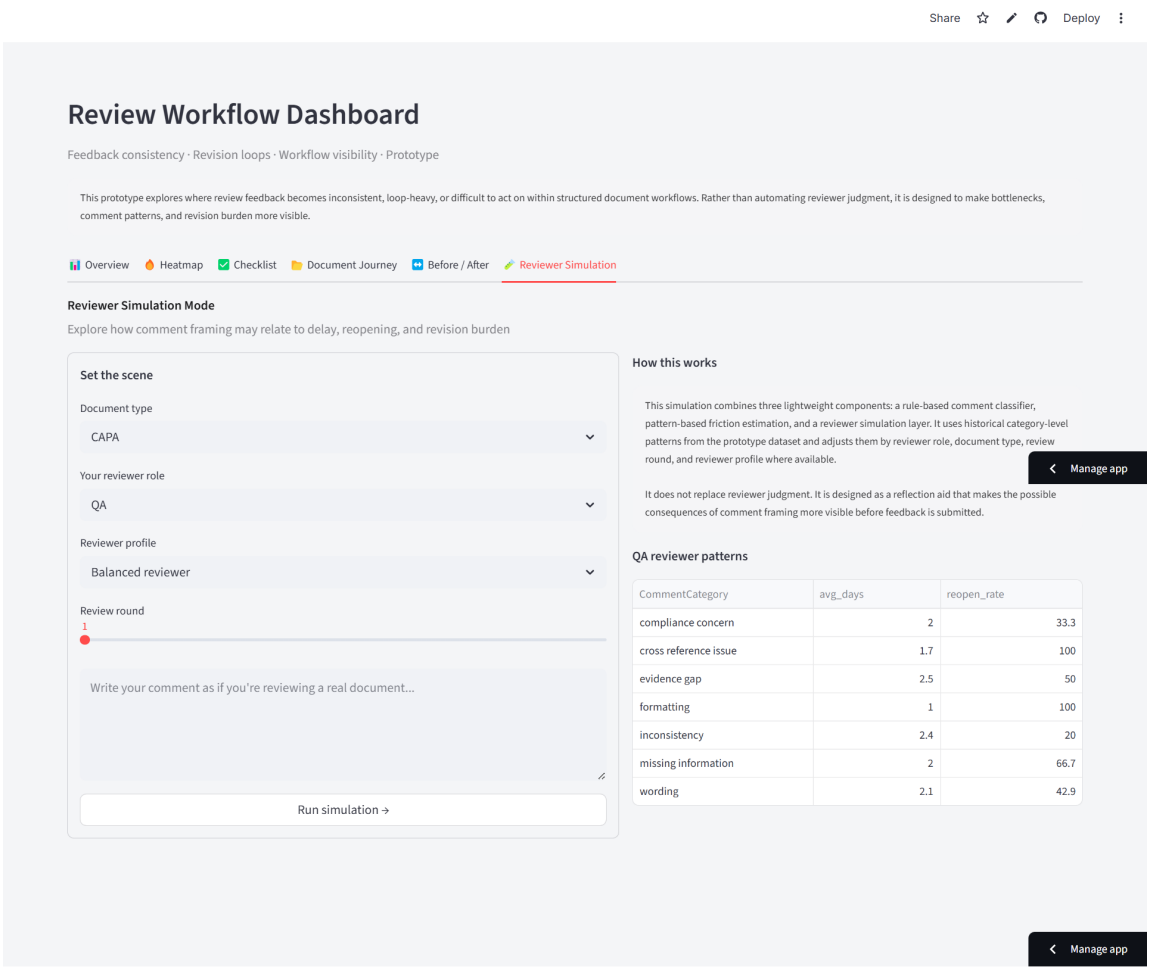


Fig. 6. Reviewer Simulation tab, showing the comment framing tool with document type, reviewer role, reviewer profile, and review round controls. The QA reviewer patterns table (right) surfaces historical category-level patterns — for example, cross-reference issues are associated with a 100% reopen rate, while inconsistency comments average 2.4 days and a 20% reopen rate.

Phase 2: Pattern Detection Layer

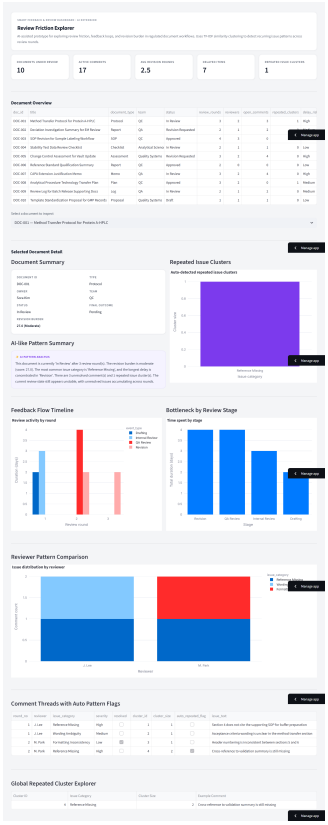


Fig. 7. Review Friction Explorer: the pattern detection layer adding TF-IDF clustering over the dashboard dataset. For a selected document (DOC-001), the layer surfaces: a document summary with revision burden score (27.0, moderate); auto-detected repeated issue clusters (Reference Missing dominates); a plain-language pattern summary describing the document's review state; a feedback flow timeline by round; bottleneck by review stage; reviewer pattern comparison; and comment threads with auto pattern flags.

The workflow dashboard addressed friction visibility at the aggregate level — surfacing patterns across documents and reviewer roles. It did not address the per-document question: for a specific document currently under review, what structural patterns have accumulated, and at which stage has friction concentrated? The TF-IDF pattern detection layer (Figure 7) addresses this residual gap as a conceptual extension of the dashboard, applying the same judgment-preservation principle to document-level pattern analysis.

The layer is organized around three functions that each address a distinct component of friction invisibility identified in the pilot evaluation.

TF-IDF clustering detects recurring patterns across comment text without manual tagging — grouping similar comments across documents and reviewers into issue clusters that surface the structural categories of friction accumulation. This addresses the opacity of individual comment threads: a reviewer whose wording comments are consistently associated with long turnaround times cannot see that pattern from within the thread itself.

Revision burden scoring quantifies the coordination load associated with individual records, combining review round count, comment volume, reopen events, and cluster concentration into a single interpretable score. This surfaces the per-document cost of friction accumulation in a form that can be compared across documents without requiring manual audit.

Pattern summaries describe each document's review state in plain language — identifying the dominant issue category, the stage at which bottlenecks have concentrated, and whether the current review state is stable or accumulating. These summaries are observability outputs, not recommendations: they describe what has happened, not what should happen next.

Throughout, the layer makes explicit what it does not do: it does not recommend actions, make accountability decisions, replace reviewer judgment, or modify validated documentation systems. This constraint is stated in the interface itself — a design choice that reflects the judgment-preservation principle governing all computational support in GMP-regulated environments [1].

Pilot Evaluation

I evaluated both components with $N = 20$ participants recruited from a pharmaceutical quality team with QA/QC review experience. This sample size is consistent with established think-aloud methodology, in which $N = 20$ expert participants in a domain-specific population reliably surfaces dominant usability and friction patterns [5]. The same-sample design was a deliberate methodological choice to eliminate between-participant variance in domain knowledge as a confound, prioritizing internal consistency over sample diversity. Recruiting from a single expert team ensured consistent baseline familiarity with GMP review conventions, though it may have inflated acceptance ratings relative to a more diverse sample — a limitation acknowledged below. The evaluation was conducted outside any live validated system, a necessary condition in GMP contexts where introducing an unvalidated interface into a production quality workflow would itself constitute a compliance risk.

Participants completed task-based evaluation scenarios using both the dashboard and the pattern detection layer, and provided feedback on purpose clarity, friction visibility, checklist usability, simulation interpretability, and adoption intent. Post-task discussion probed the conceptual accessibility of the friction-visibility framing.

4 Findings

Friction Visibility as Fairness Infrastructure

The most consequential participant response was not about interface usability but about what visibility of review burden would change organizationally. All 20 participants anticipated reduced accountability asymmetry from the dashboard — a finding that emerged consistently across both submitter and reviewer roles without prompting. Participants described a recurring condition in current practice: review delays accumulate, documents circulate without clear progress signals, and when closure finally fails or slows, responsibility tends to be assigned without access to the coordination conditions that produced the outcome. As one participant put it: "the delay gets attributed to whoever submitted the document, not to the review structure that produced it." Another described the anticipated value more directly: "this would make it harder to blame the person at the end of the queue when the problem was in the middle of the process."

What this response makes visible is a shift in interpretation. When coordination patterns were not visible, participants described delay in outcome terms; when friction patterns were surfaced, delay was reinterpreted as emerging from structural accumulation across the workflow. Rather than eliminating responsibility, visibility redistributed it:

responsibility became tied to observable coordination conditions rather than inferred outcomes. In this sense, friction visibility functioned not only as workflow support but as fairness infrastructure.

Role-Differentiated Utility

Participants described the dashboard's value differently depending on their role in the review process. From the perspective of document submitters, the primary value was visibility into where their document stood and what type of review activity was generating delay. From the perspective of QA reviewers, the value was different: the dashboard offered a way to monitor and regulate their own review workload across documents. All 20 participants expressed adoption intent, with this role-differentiated framing emerging as the dominant account of why the system would be useful.

The Checklist vs. Pattern Detection Dissociation

The high usability ratings observed across both components — including a mean of 5.0/5 for the pattern detection layer — are better understood as a domain effect than a measurement artifact. GMP review environments have no existing analogue for external friction observability tools; participants evaluated both components not against competing systems but against the absence of any structured visibility support. The ceiling reflects the size of the gap the observability layer fills rather than the insensitivity of the measure. The TF-IDF clustering layer received a final usability rating of 5.0/5 and was endorsed by all 20 participants as useful once demonstrated — yet it required initial orientation to interpret in a way that the checklist did not. The checklist was immediately accessible because it addresses a problem practitioners already recognize — inconsistent review preparation. The pattern detection layer required initial orientation because it addresses a problem practitioners have not previously had the tools to name — structural friction accumulation.

Crucially, this initial unfamiliarity did not produce rejection. Once the framing was explained, all 20 participants endorsed the concept as genuinely useful. As one participant noted: "I never thought of revision loops as something you could measure. But once you see it that way, it's obvious that some documents are just structurally harder." This dissociation is the study's central empirical finding: usefulness and immediate legibility are not the same thing. A tool can be highly useful for a problem that practitioners have not yet learned to see. This implies that the appropriate metric for evaluating an accountability-critical design intervention is not whether practitioners immediately recognize its value, but whether they endorse it once the problem it addresses has been made legible — and that the interpretive overhead of problem-naming is itself a design responsibility, not a user limitation.

The Organizational Adoption Problem

All 20 participants expressed adoption intent while simultaneously identifying a structural barrier to deployment: whether regulated organizations would adopt a system that makes review reasoning more transparent. Participants did not object to the system making review work more thorough — they objected to it making review work more visible. When friction patterns are documented, they produce a record of the coordination conditions that were present and not addressed, which in a regulatory environment subject to audit may surface accountability conditions that current practice leaves invisible.

5 Discussion

Observability as Interaction Design

The central contribution of this work is a reframing of workflow support in regulated environments. Rather than optimizing coordination, the design problem is to make coordination conditions visible. Observability is not a secondary feature of workflow systems; it is part of the interaction conditions under which review work becomes interpretable.

When coordination work is invisible, participants act under conditions of partial visibility and attribute outcomes based on incomplete information. By making coordination patterns visible, the observability layer reshapes how workflow outcomes are interpreted without automating those outcomes. This aligns with Suchman's (1987) account of situated action: action is shaped by what is visible and interpretable at the moment it is taken. Applied to document review, this means that accountability is shaped not only by what reviewers decide, but by what the system makes available for those decisions to be understood against.

Invisible Work and the Organizational Stakes of Legibility

The role-differentiated responses documented in the findings reflect the asymmetric distribution of invisible work that Star and Strauss [6] describe. The dashboard addresses both forms of invisibility in the review workflow: for submitters, it makes the accumulated review burden on their documents legible; for QA reviewers, it provides a workload monitoring function that current systems do not support. That all 20 participants across both roles reported adoption intent suggests that the observability layer is not a tool for one side of a coordination asymmetry, but a shared infrastructure for making that asymmetry discussable.

Judgment-Preserving Design in Compliance-Constrained Environments

The design constraint that governed both components — that computational outputs must support human judgment rather than substitute for it — is not merely a normative preference. In GMP environments, quality decisions must be made and documented by qualified human reviewers; any system that automates or pre-empts those decisions introduces a compliance risk. The pattern detection layer's approach — surfacing TF-IDF clusters and burden scores as observability signals rather than recommendations — is one instantiation of the judgment-preservation principle [1]. The finding that clustering outputs required initial interpretive orientation, despite ultimately receiving 5.0/5 ratings from all 20 participants, points to a productive tension: the more a system preserves human judgment by withholding recommendations, the more it requires users to bring their own interpretive frame to the output.

The Checklist as a Bridge

The checklist's immediate legibility relative to the pattern detection layer suggests a practical adoption pathway: checklist-first deployment that establishes the dashboard's credibility within existing review practice, followed by progressive introduction of pattern detection outputs as users develop the interpretive frame to use them. The conceptual barrier to friction visibility is real but not insurmountable — all 20 participants endorsed the concept once demonstrated.

More broadly, the dissociation between immediate legibility and final endorsement points to a design principle that extends beyond adoption sequencing. In accountability-critical environments, tools that address structurally invisible problems will routinely require practitioners to first acquire the conceptual vocabulary to name those problems — and this requirement is not a usability failure but a feature of the design challenge itself. The appropriate design response is not to make such tools more immediately legible by reducing their ambition, but to build the interpretive bridge

explicitly: naming the problem the tool addresses, before asking practitioners to evaluate whether the tool addresses it well. The checklist functions as precisely this kind of bridge — not only as a simpler entry point, but as the instrument through which practitioners develop the problem category that makes the pattern detection layer's value legible.

Limitations

Several limitations bound the interpretation of these findings. The synthetic dataset underrepresents the terminological and structural variance of live review records, which may have reduced the discriminability of clustering outputs and contributed to the high acceptance ratings. Participants were recruited from a single team, limiting generalizability. The study does not establish effects on formal quality metrics — CAPA closure time, deviation recurrence rates, or audit findings.

6 Conclusion

This paper introduced review friction visibility as a design problem in regulated document environments, and demonstrated that an external observability layer can surface hidden coordination work without modifying validated documentation systems or automating quality decisions. More specifically, it showed that observability does not eliminate friction; it makes friction legible as an interaction condition that shapes accountability before failure occurs. All 20 participants rated both components 5.0/5, expressed adoption intent, and anticipated reduced accountability asymmetry from friction visibility.

The study's central empirical finding is a dissociation between immediate legibility and final endorsement: the checklist was immediately accessible because it addresses a problem practitioners already recognize, while the pattern detection layer required initial conceptual orientation because it addresses a problem practitioners have not previously had the tools to name. Both arrived at equivalent ratings. The interpretive overhead of the pattern detection layer is not a usability failure — it is a signal that making structural friction visible requires building a new problem category, not just a new interface.

Three contributions follow. First, review friction visibility is introduced as a design problem specific to regulated document environments, where the judgment-preservation constraint uniquely restricts what computational support is permissible. Second, the external metadata layer approach is demonstrated as a viable design strategy for surfacing coordination patterns within compliance boundaries. Third, the theoretical framing — coordination work as an upstream condition of accountability assignment — connects HCI concerns about workflow visibility to broader organizational questions about how responsibility is structured before formal judgment begins.

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